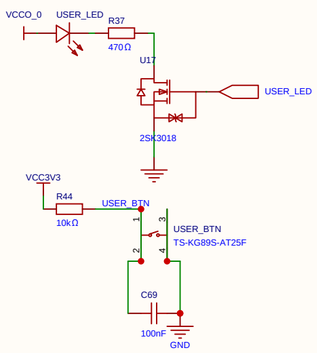
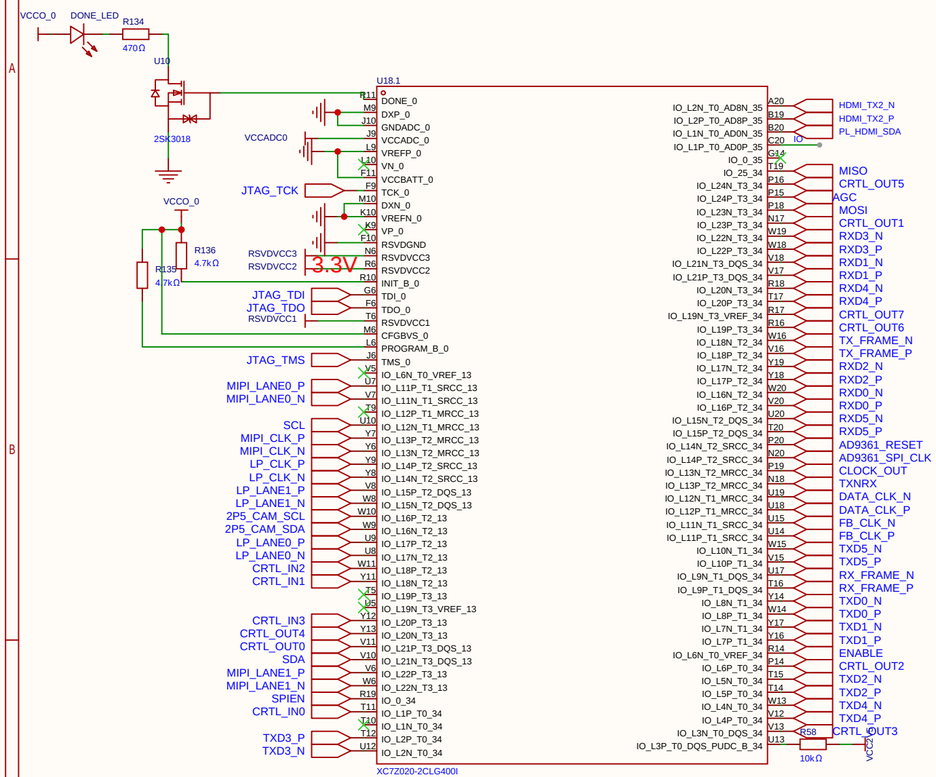
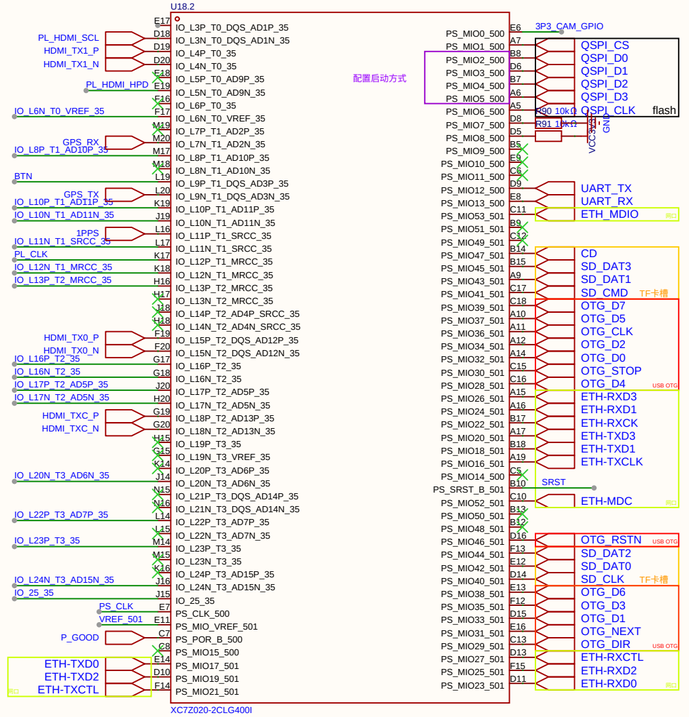
TODO:
1.修改UART线序（接反了）
2.修改Flash线序（接反了）
3.螺丝接地、开窗屏蔽
4.修改射频头接口
5.可能考虑不要NCP59801CMLADJTCG，不好买

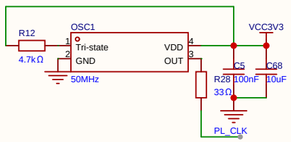
未跑通：
1.USB3320的芯片ID识别不到，怀疑焊接问题
2.Hdmi（没进行测试）
3.MIPI（没进行测试）
4.GPS（没进行测试）
5.TFCARD（没进行测试）

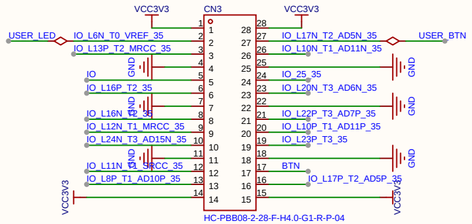
原理图	FreeSdr_sch	创建日期	2026-09-17
		更新日期	2026-10-31
板子	FreeSdr	图页	TODO
绘制		FreeSdr	
审阅			
		版本	尺寸	页 1 共 11
嘉立创EDA		V1.0	A4	嘉立创EDA

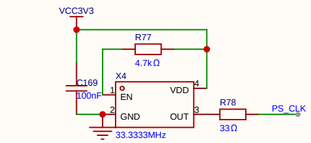


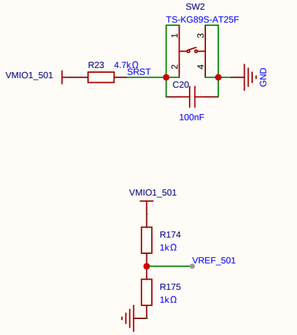




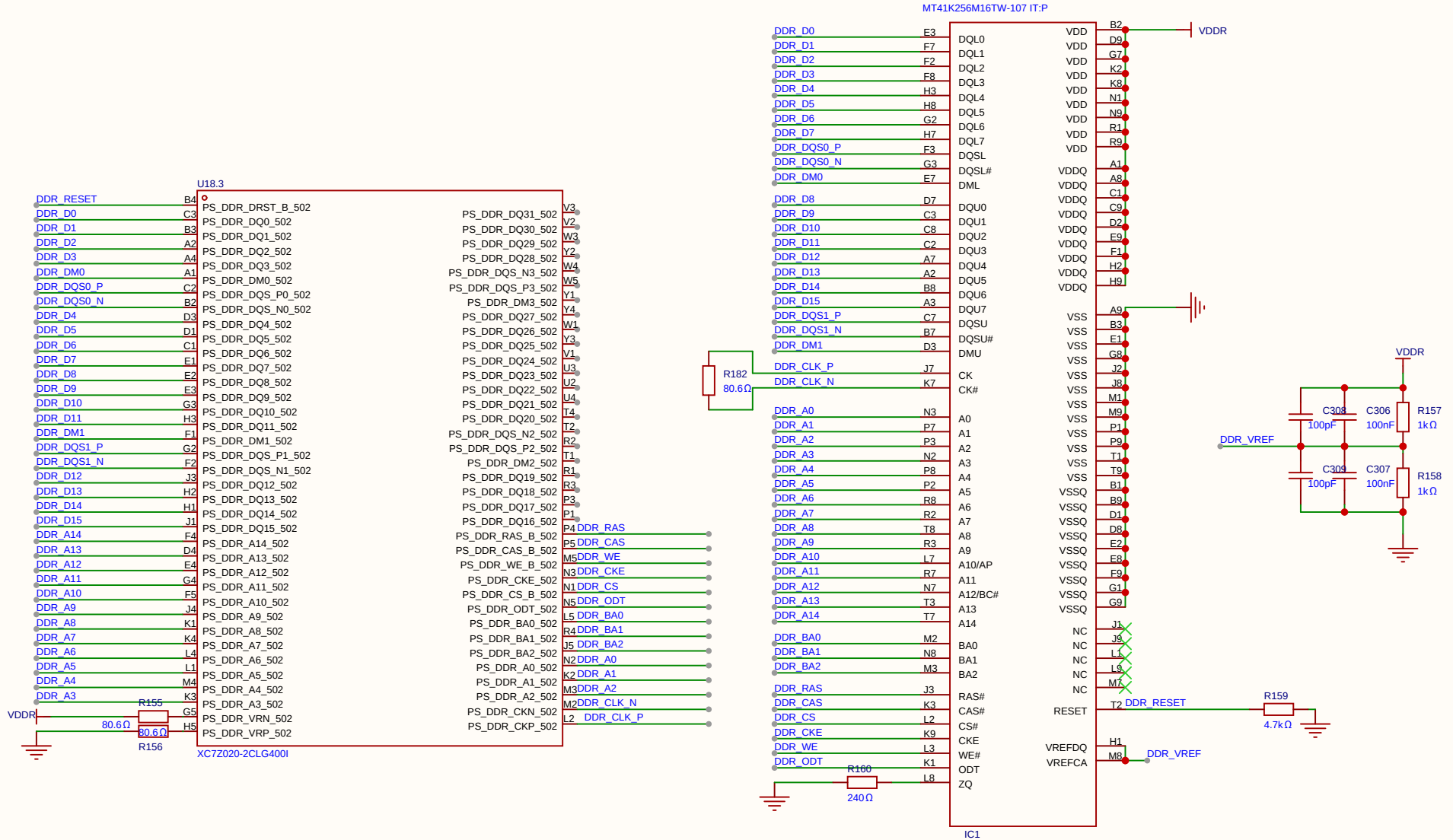




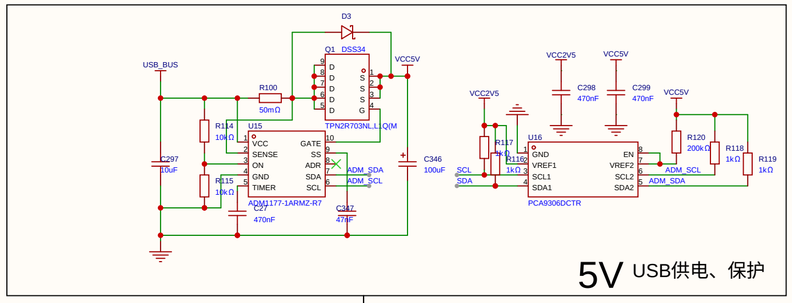


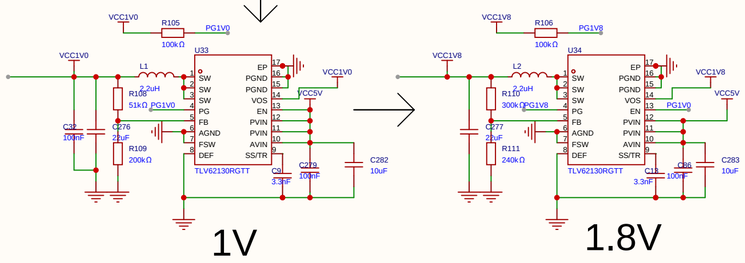


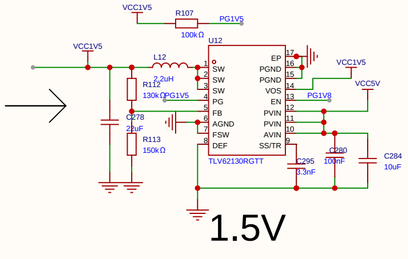
原理图	FreeSdr_sch	更新日期	2026-10-31
图页	XCz7020	创建日期	2026-10-17
绘制		物料编码	
审阅		FreeSdr	
		版本	尺寸
嘉立创EDA		V1.0	A4
		页 2 共 11	
		嘉立创EDA	

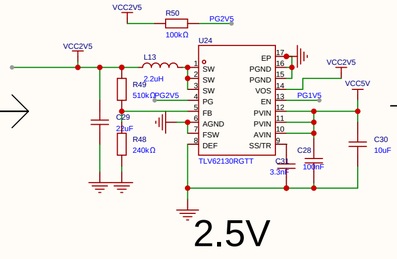

U18.3
XC7Z020-2CLG400I
DDR_RESET
DDR_D0
DDR_D1
DDR_D2
DDR_D3
DDR_DM0
DDR_DQS0_P
DDR_DQS0_N
DDR_D4
DDR_D5
DDR_D6
DDR_D7
DDR_D8
DDR_D9
DDR_D10
DDR_D11
DDR_DM1
DDR_DQS1_P
DDR_DQS1_N
DDR_D12
DDR_D13
DDR_D14
DDR_D15
DDR_A14
DDR_A13
DDR_A12
DDR_A11
DDR_A10
DDR_A9
DDR_A8
DDR_A7
DDR_A6
DDR_A5
DDR_A4
DDR_A3
PS_DDR_DRST_B_502
PS_DDR_DQ0_502
PS_DDR_DQ1_502
PS_DDR_DQ2_502
PS_DDR_DQ3_502
PS_DDR_DM0_502
PS_DDR_DQS_P0_502
PS_DDR_DQS_N0_502
PS_DDR_DQ4_502
PS_DDR_DQ5_502
PS_DDR_DQ6_502
PS_DDR_DQ7_502
PS_DDR_DQ8_502
PS_DDR_DQ9_502
PS_DDR_DQ10_502
PS_DDR_DQ11_502
PS_DDR_DM1_502
PS_DDR_DQS_P1_502
PS_DDR_DQS_N1_502
PS_DDR_DQ12_502
PS_DDR_DQ13_502
PS_DDR_DQ14_502
PS_DDR_DQ15_502
PS_DDR_A14_502
PS_DDR_A13_502
PS_DDR_A12_502
PS_DDR_A11_502
PS_DDR_A10_502
PS_DDR_A9_502
PS_DDR_A8_502
PS_DDR_A7_502
PS_DDR_A6_502
PS_DDR_A5_502
PS_DDR_A4_502
PS_DDR_A3_502
PS_DDR_VRN_502
PS_DDR_VRP_502
PS_DDR_DQ31_502
PS_DDR_DQ30_502
PS_DDR_DQ29_502
PS_DDR_DQ28_502
PS_DDR_DQS_N3_502
PS_DDR_DQS_P3_502
PS_DDR_DM3_502
PS_DDR_DQ27_502
PS_DDR_DQ26_502
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PS_DDR_DQ24_502
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PS_DDR_DQ22_502
PS_DDR_DQ21_502
PS_DDR_DQ20_502
PS_DDR_DQS_N2_502
PS_DDR_DQS_P2_502
PS_DDR_DM2_502
PS_DDR_DQ19_502
PS_DDR_DQ18_502
PS_DDR_DQ17_502
PS_DDR_DQ16_502
PS_DDR_RAS_B_502
PS_DDR_CAS_B_502
PS_DDR_WE_B_502
PS_DDR_CKE_502
PS_DDR_CS_B_502
PS_DDR_ODT_502
PS_DDR_BA0_502
PS_DDR_BA1_502
PS_DDR_BA2_502
PS_DDR_A0_502
PS_DDR_A1_502
PS_DDR_A2_502
PS_DDR_CKN_502
PS_DDR_CKP_502
DDR_RAS
DDR_CAS
DDR_WE
DDR_CKE
DDR_CS
DDR_ODT
DDR_BA0
DDR_BA1
DDR_BA2
DDR_A0
DDR_A1
DDR_A2
DDR_CLK_N
DDR_CLK_P
VDDR
R155
80.6Ω
R156
80.6Ω
MT41K256M16TW-107 IT:P
IC1
R182
80.6Ω
R160
240Ω
R159
4.7kΩ
DDR_RESET
DDR_VREF
C308
100pF
C306
100nF
C309
100pF
C307
100nF
R157
1kΩ
R158
1kΩ
VDDR
DDR_VREF

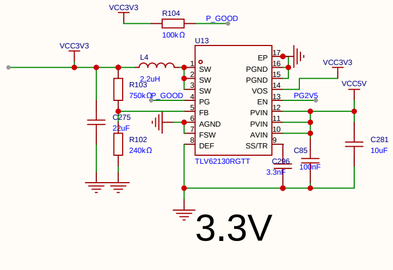
5V->1V->1.8V->1.5V->3.3V->2.5V ->1.2V

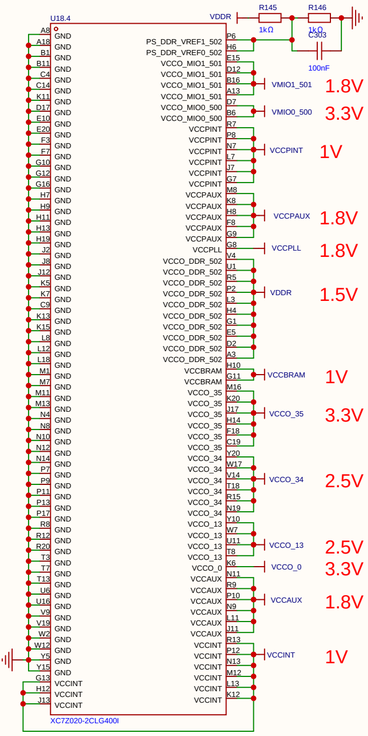


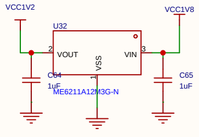


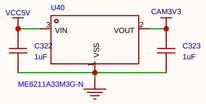


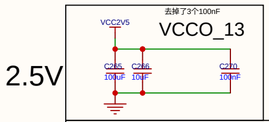




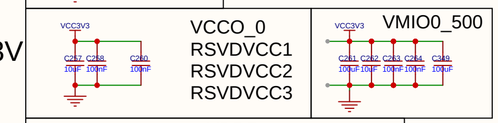




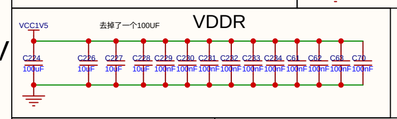




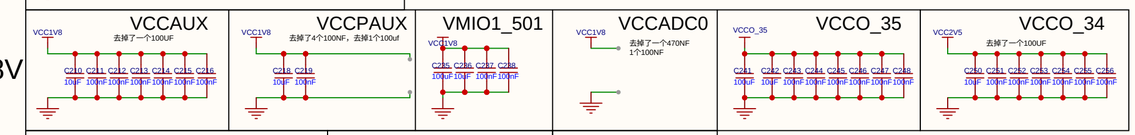
2.5V



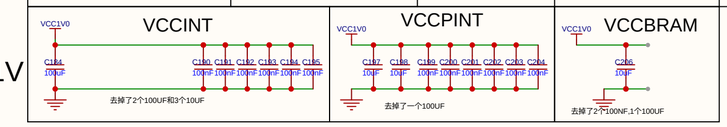
3.3V



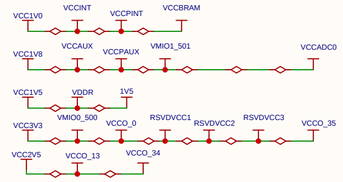
1.5V

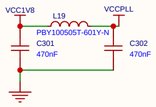


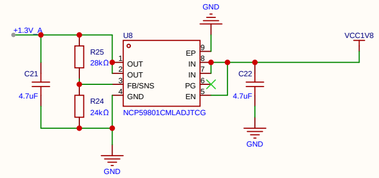
1.8V



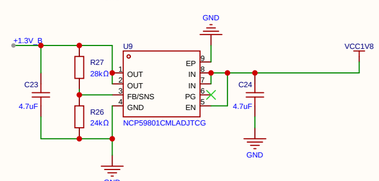
1V







1.3V_A

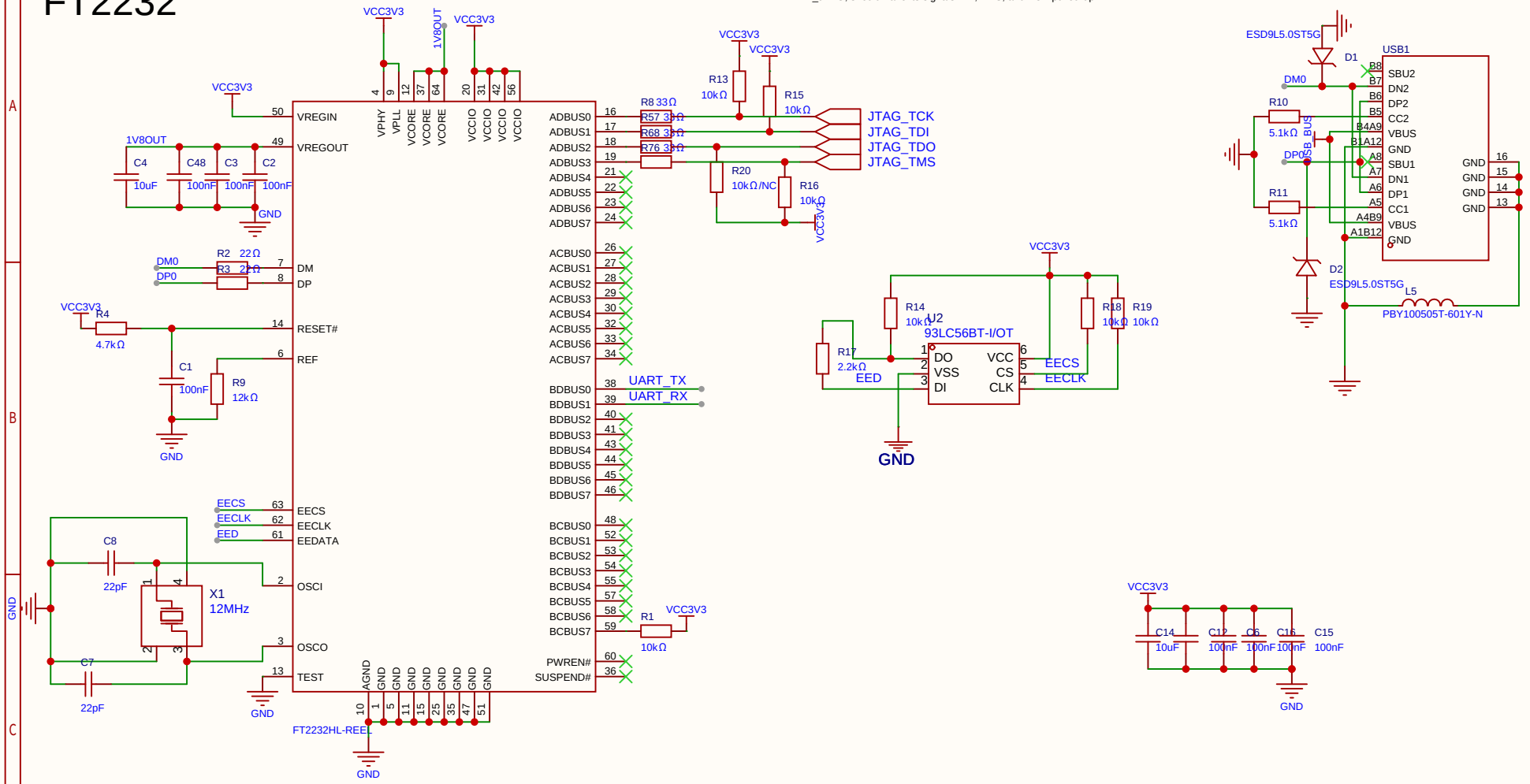


1.3V_B

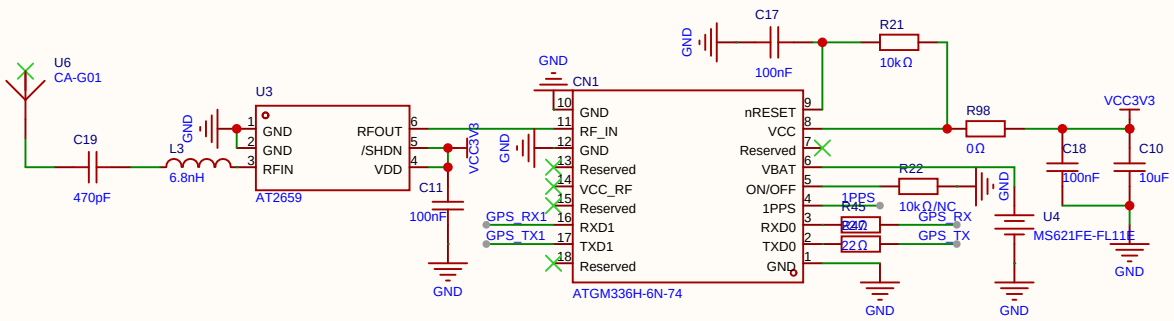
原理图	FreeSdr_sch	更新日期	2026-10-31
		创建日期	2026-10-10
图页	POWER DONE	物料编码	
绘制		FreeSdr	
审阅			
		版本	V1.0
		尺寸	A4
		页 4 共 11	
嘉立创EDA		嘉立创EDA	

FT2232

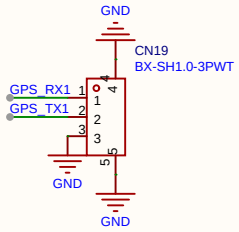
The PL system JTAG interface,
PL_JTAG, should have its signals TDI, TMS, and TCK pulled-up.

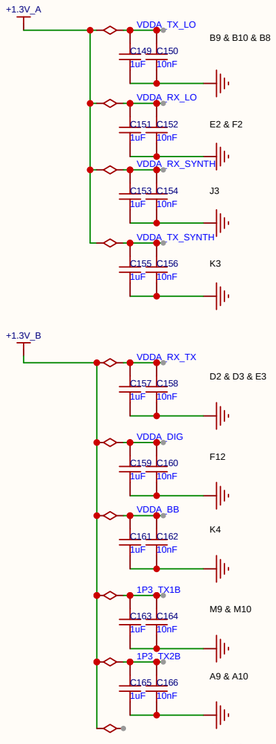


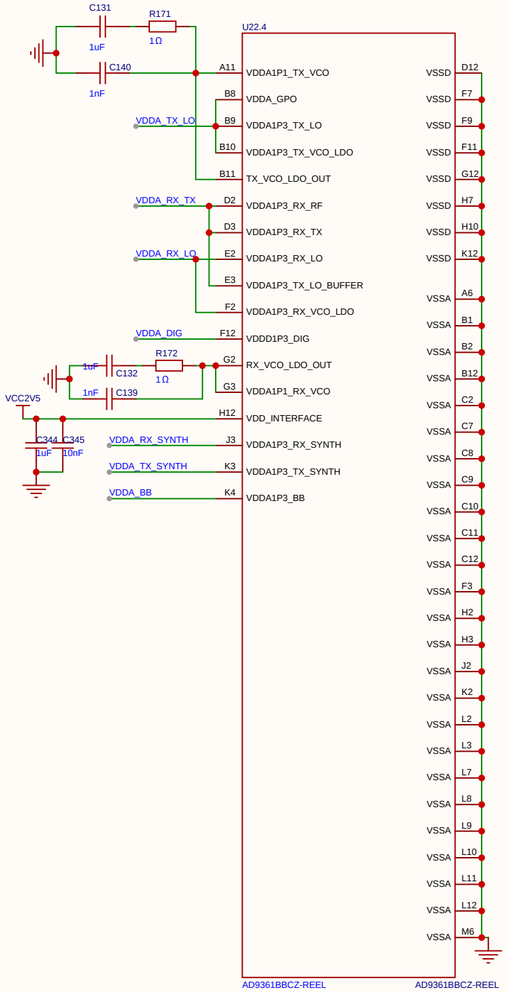
修改记录：
修改BDBUS7为上拉
增加Jtag匹配电阻，默认上拉电平
移除Jtag、UART的ESD

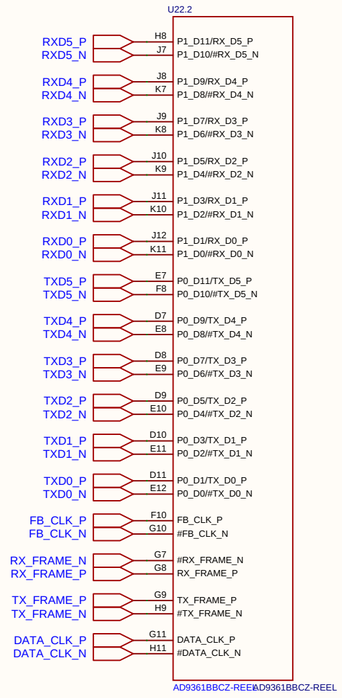


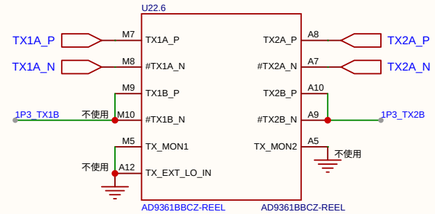
5号管脚拉低为关闭模块，悬空即可

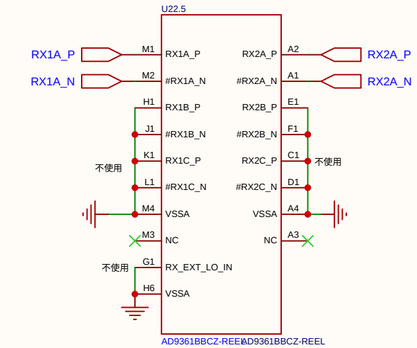


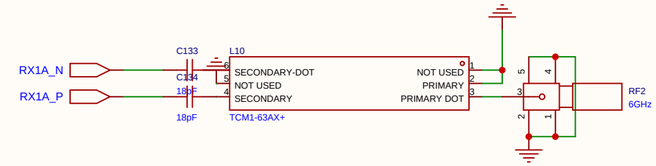


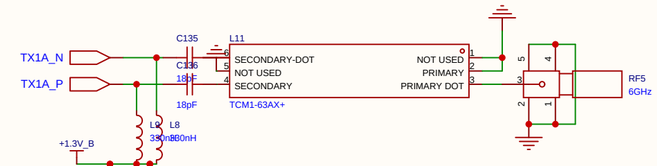


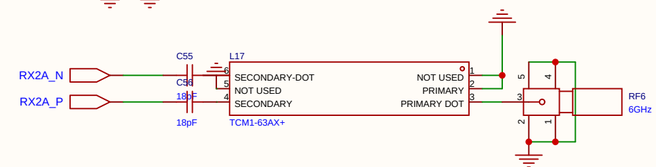


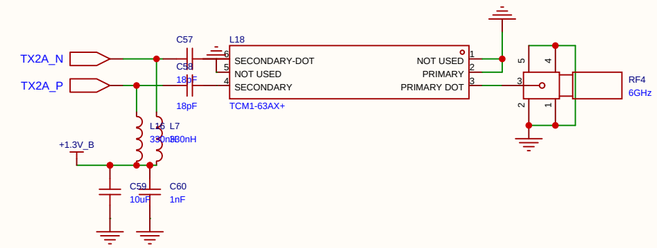


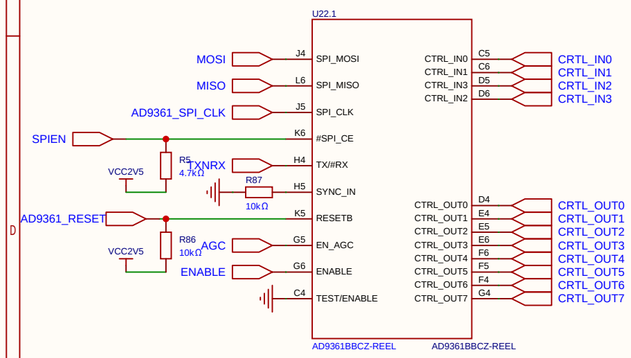




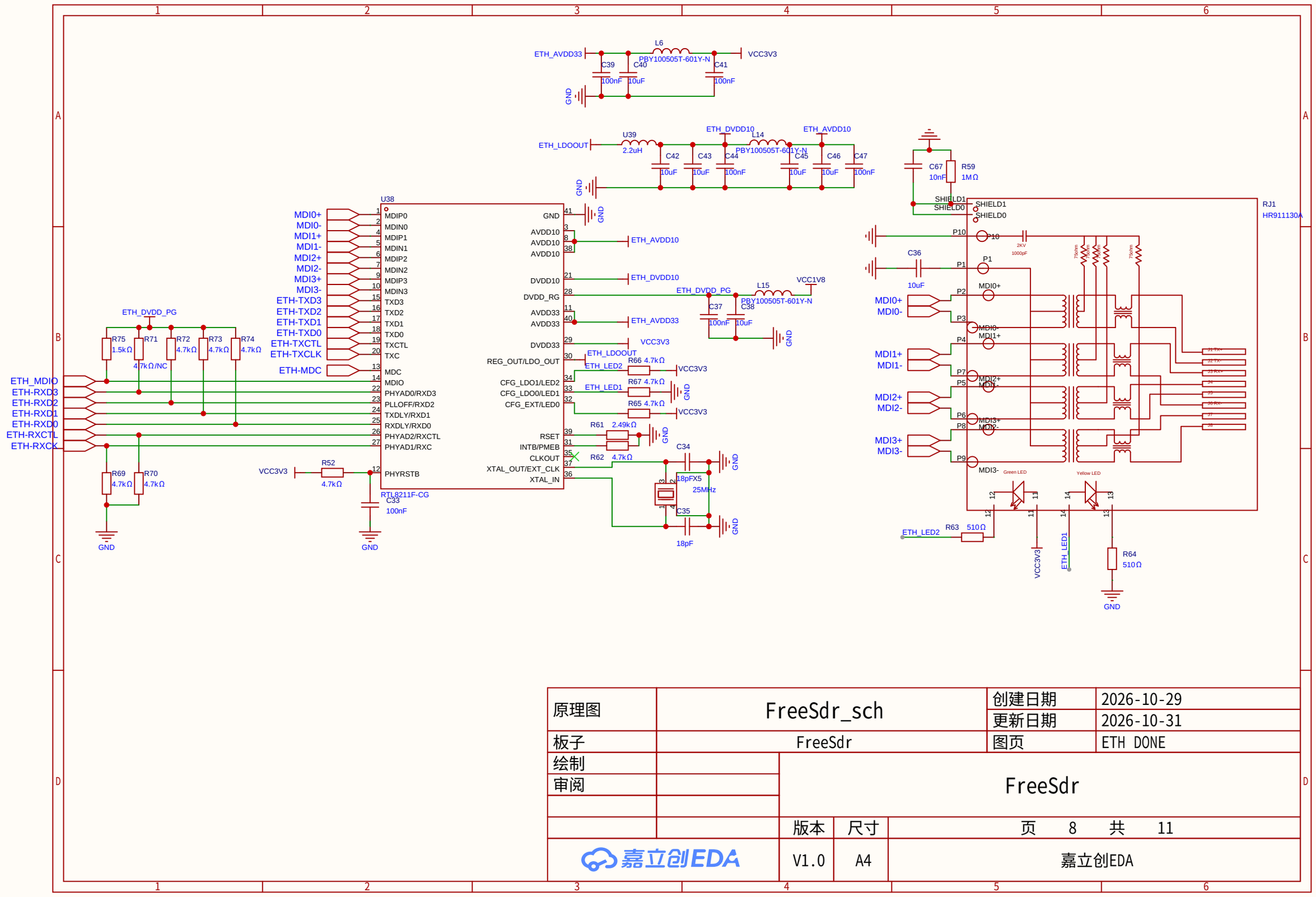








原理图	FreeSdr_sch	创建日期	2026-09-20
		更新日期	2026-10-31
板子	FreeSdr	图页	AD9361 DONE
绘制		FreeSdr	
审阅			
		版本	尺寸
嘉立创EDA		V1.0	A4
		页 7 共 11	
		嘉立创EDA	

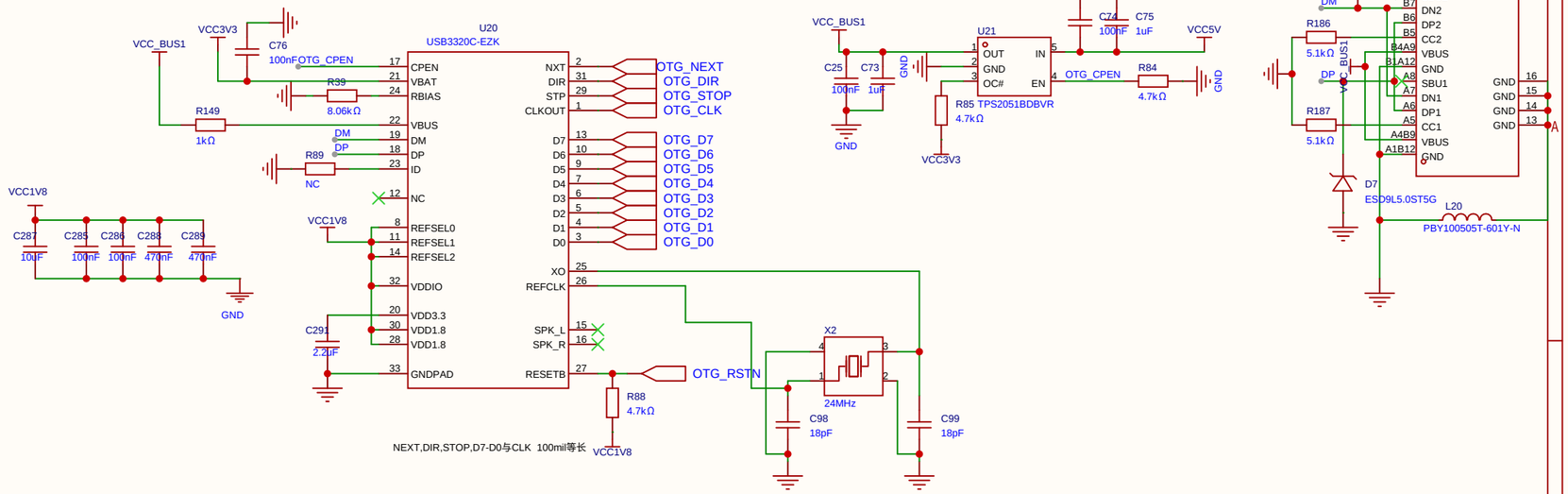


原理图	FreeSdr_sch	创建日期	2026-10-29
		更新日期	2026-10-31
板子	FreeSdr	图页	ETH DONE
绘制		FreeSdr	
审阅			
		版本	尺寸	页 8 共 11
嘉立创EDA		V1.0	A4	嘉立创EDA

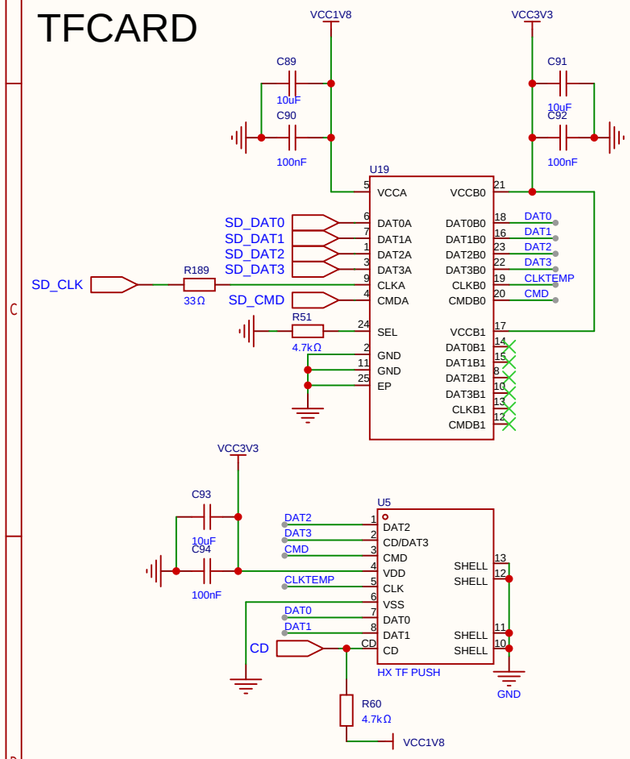
烧录/串口

值得注意的JTAG在BANK0，电压为3.3V
串口在BANK500，电压为3.3V
但是PS端的RST在BANK501，供电为1.8V
如果要使用，需要使用外部电压转换芯片
NXS0101GMX

OTG



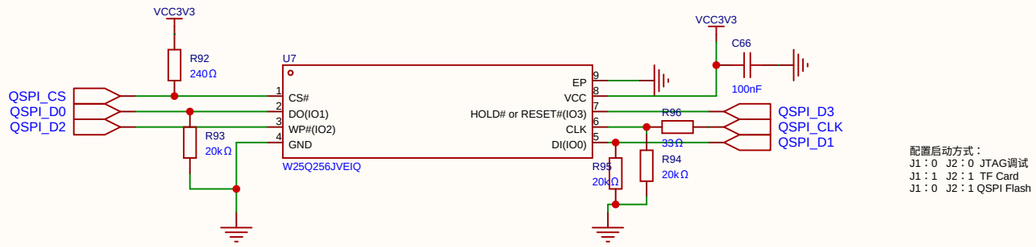
TFCARD



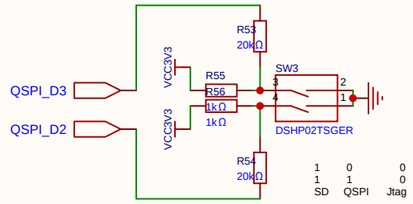
没有WP，选用SD0，开启CD即可

200mil等长

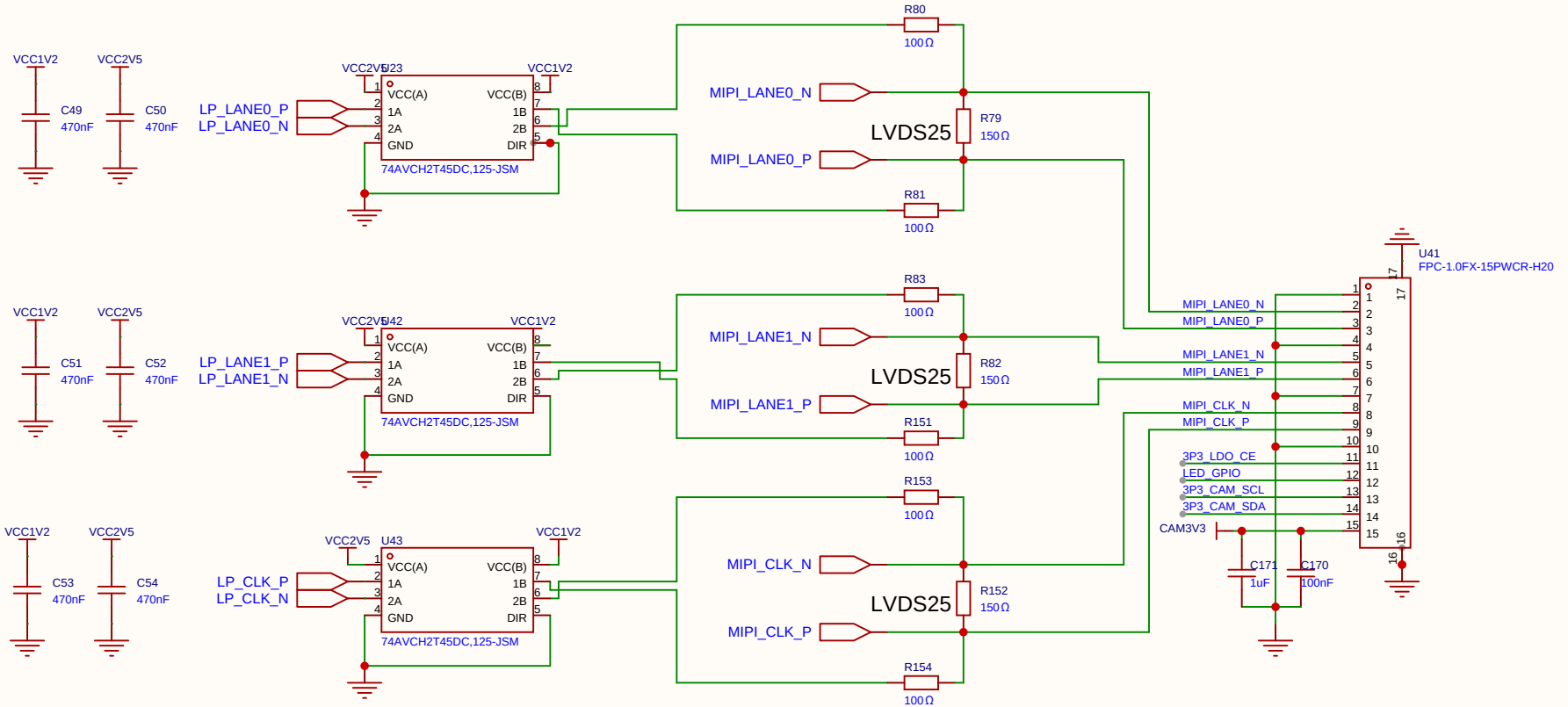
Flash



配置启动方式：
J1：0 J2：0 JTAG调试
J1：1 J2：1 TF Card
J1：0 J2：1 QSPI Flash

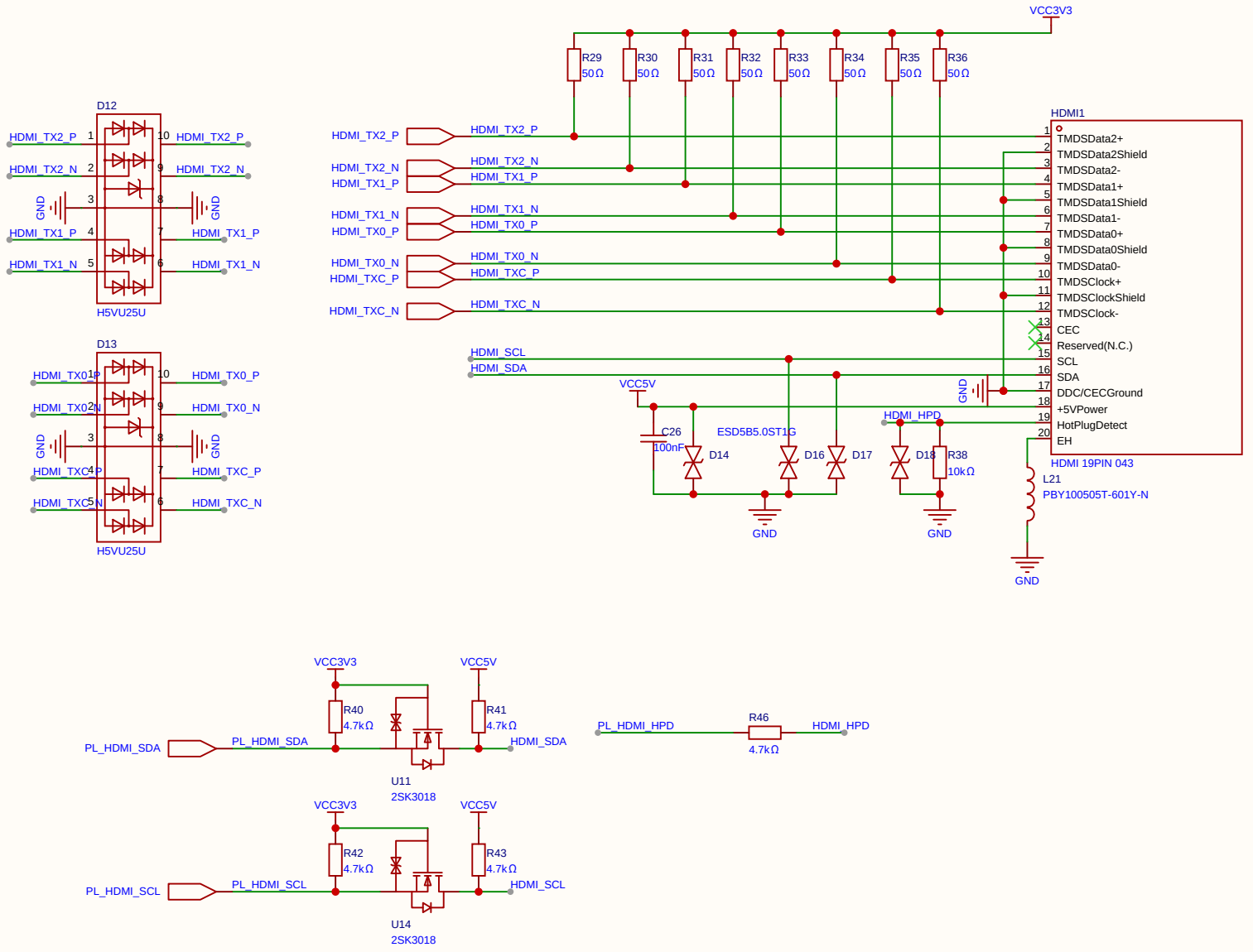


原理图	FreeSdr_sch	创建日期	2026-10-10
		更新日期	2026-10-31
板子	FreeSdr	图页	Peripheral DONE
绘制		FreeSdr	
审阅			
		版本	尺寸	页 9 共 11
嘉立创EDA		V1.0	A4	嘉立创EDA



因为我的摄像头上有晶振，所以不需要给摄像头提供时钟了

原理图	FreeSdr_sch	创建日期	2026-10-03
		更新日期	2026-10-31
板子	FreeSdr	图页	MIPI DONE
绘制		FreeSdr	
审阅			
		版本	尺寸	页 10 共 11
嘉立创EDA		V1.0	A4	嘉立创EDA


VCC3V3
R29 50Ω
R30 50Ω
R31 50Ω
R32 50Ω
R33 50Ω
R34 50Ω
R35 50Ω
R36 50Ω
D12
HDMI_TX2_P
HDMI_TX2_N
GND
HDMI_TX1_P
HDMI_TX1_N
H5VU25U
D13
HDMI_TX0_P
HDMI_TX0_N
HDMI_TXC_P
HDMI_TXC_N
HDMI_SCL
HDMI_SDA
HDMI1
TMDSData2+
TMDSData2Shield
TMDSData2-
TMDSData1+
TMDSData1Shield
TMDSData1-
TMDSData0+
TMDSData0Shield
TMDSData0-
TMDSClock+
TMDSClockShield
TMDSClock-
CEC
Reserved(N.C.)
SCL
SDA
DDC/CECGround
+5VPower
HotPlugDetect
EH
HDMI 19PIN 043
VCC5V
C26
100nF
ESD5B5.0ST1G
D14
D16
D17
D18
R38
10kΩ
HDMI_HPD
L21
PBY100505T-601Y-N
GND
R40
4.7kΩ
R41
4.7kΩ
PL_HDMI_SDA
U11
2SK3018
R42
4.7kΩ
R43
4.7kΩ
PL_HDMI_SCL
U14
2SK3018
R46
4.7kΩ
PL_HDMI_HPD